DO NOW — RECALL

# Lesson 052 · Binary Search Trees with Node Objects

Name: _____ Class: _____ Date: _____

**1. What does LinkedList<T>.AddAfter use to place a new node?**

**2. What does HEAD mean for a linked list?**

**3. What is an invariant of a BST?**

**4. A binary search tree has root 10, left child 6 and right child 18. Insert 15, then 12, using the usual smaller-left, larger-right rule. Determine the parent and side of the new node 12.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

An existing node reference.

### 2. Model answer

Its first node.

### 3. Model answer

Left subtree keys less and right keys greater under chosen policy.

### 4. Model answer

12 is the left child of 15. Its search path is right from 10, left from 18, then left from 15.

### By the end, you can…

• A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

• Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

• Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

### Vocabulary — know these terms

BST — Binary search tree ordered by key.

root — Topmost tree node.

leaf — Node without child nodes.

inorder — Left subtree, root, right subtree.

NEW LEARNING

# Read, annotate and explain

## Discover

A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

## Trace

Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

## Evaluate

Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Insert key 7 into BST root 8, left 3, right 10, with 6 right of 3.

### Explain the trace

7 becomes right child of 6; follow comparisons until a null child is reached.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

BST root8, left3, right10, insert 6. Show comparisons and location. [6]

### Question 2

For BST with keys 8,3,10,6, give inorder traversal and explain sorting. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why BST search may be O(log n) when balanced but O(n) when skewed. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An unbalanced BST is guaranteed to have logarithmic search time.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.